Assignment Sheet

Task 1 – Number conversion (2+2 Points)

Convert the following hexadecimal numbers into their octal representation by firstly converting them to unsigned binary. For each of your (interim) results, use the minimum number of digits and clearly indicate to which number system it belongs.

a) $(F17)_{16}$

b) $(C1A)_{16}$

Task 2 – Two's complement (3+3 Points)

Convert the following binary numbers in two's complement representation into their corresponding decimal representations. Make sure to include all necessary conversion steps.

a) $(01010101)_2$

b) $(11110000)_2$

Task 3 – Positional Number Systems (2+6 Points)

Answer the following questions for the number system with base $b = 3$

a) What digits (symbols) does the number system use?

b) What is the representation of $(33)_{10}$ in the given system? Use the minimum number of digits necessary (unsigned representation).

Task 4 – Signed vs. Unsigned Interpretation (3+3+3 Points)

The 8-bit binary number $(11110101)_2$ is given.

a) What is its decimal value if interpreted as unsigned?

b) What is its value if interpreted as a signed two's complement number?

c) Why are the results different?

Task 5 – Integer overflow (3 Points)

Explain the difference between unsigned and signed integer overflow. Why is it so important to detect overflow?

This assignment is due on 07.11.2025 at 10 am.